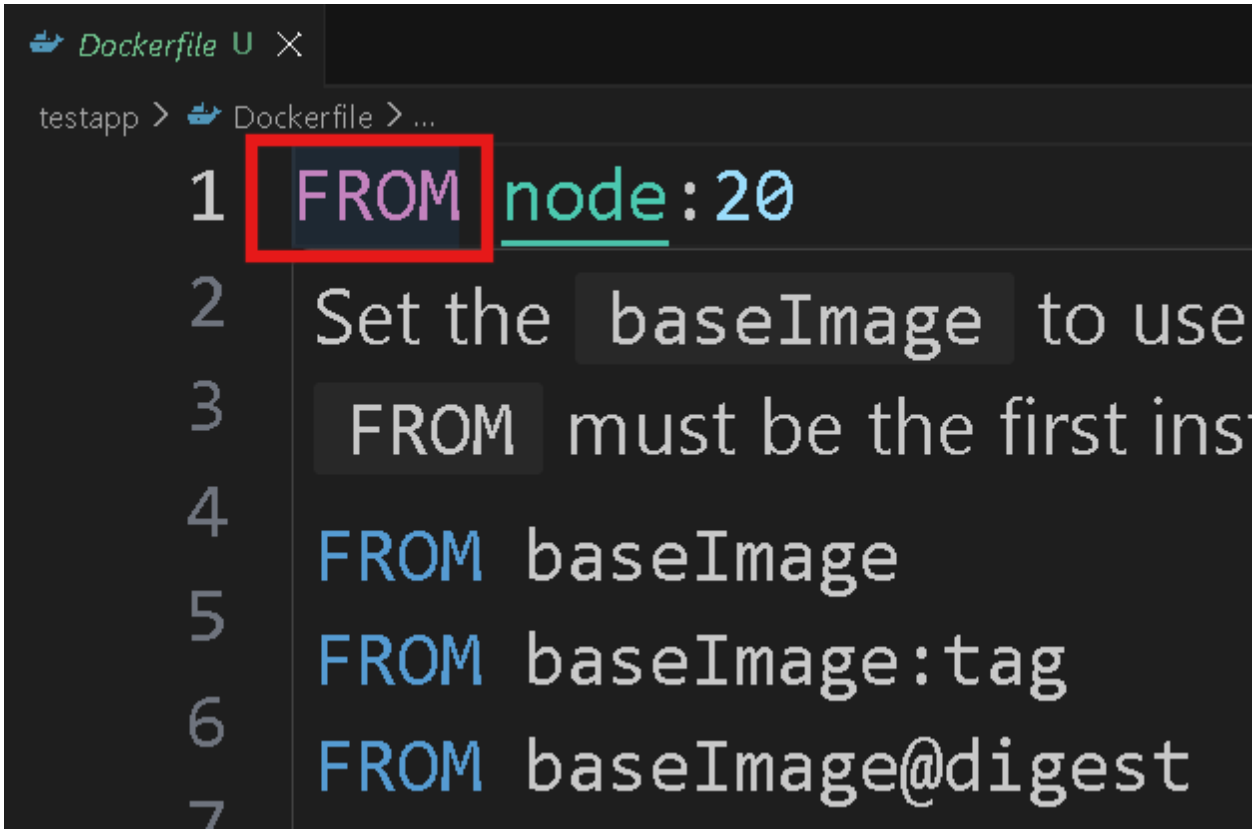
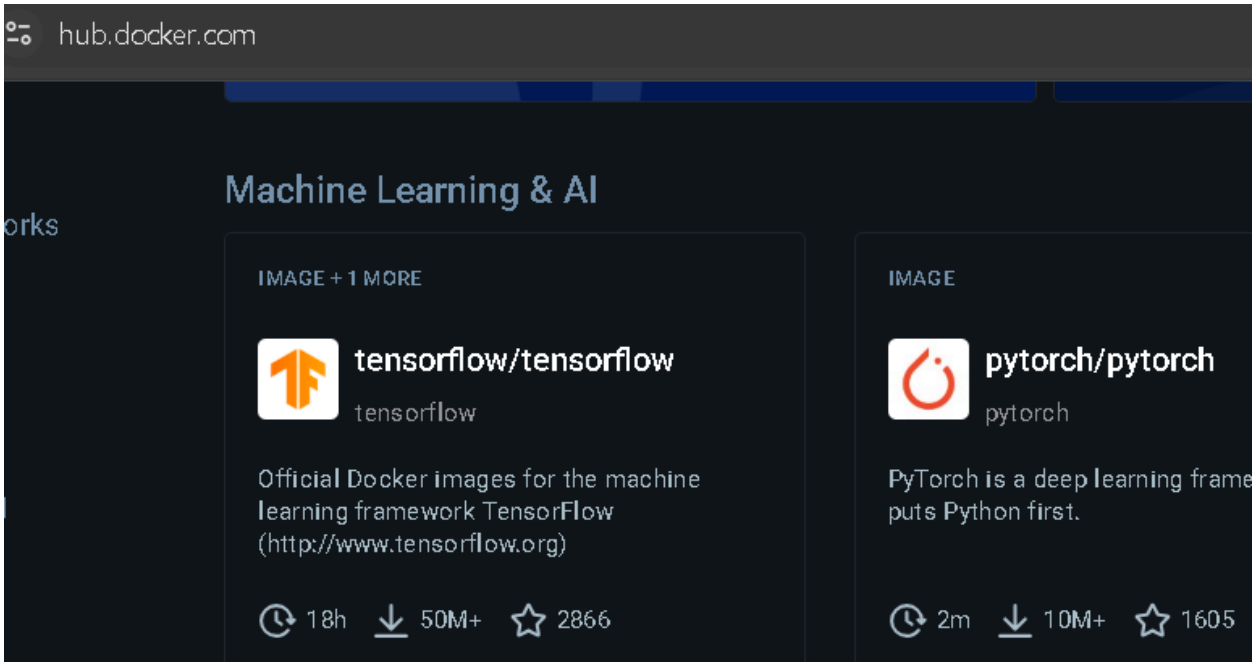
Pre-defined Docker Image:

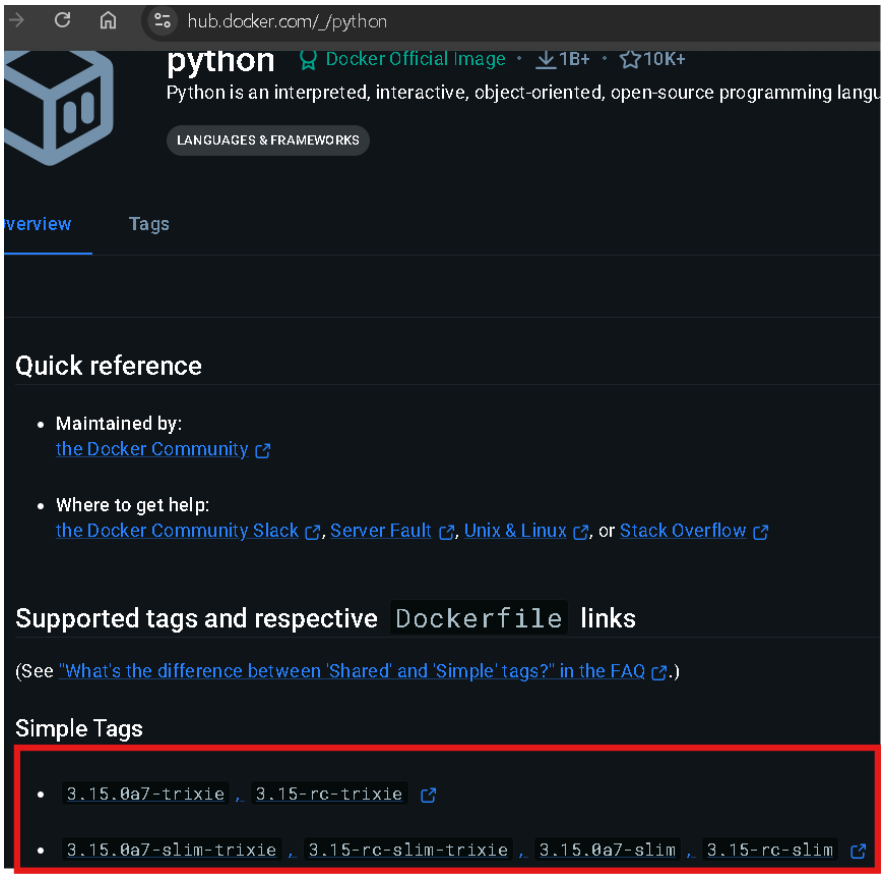
With the help of docker, we have created our personal image/customized image,
When you hover over FROM,



Which shows baseImage which points towards node:20
In the dockerhub.com, which has pre-defined images provided by the big companies.
Which are most commonly used images,



For python, which has many versions,



```
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker pull python
Using default tag: latest
latest: Pulling from library/python
21d10573d00c: Pulling fs layer
60f77666b15c: Pull complete
8f6ad858d0a4: Pull complete
075ad1518904: Pull complete
ee3a0e7d77f0: Pull complete
8688d0f2f567: Pull complete
b012eb15dff0: Pull complete
784a92c39bcf: Download complete
e508803cca92: Download complete
Digest: sha256:0040600478fe761fbaea26b16cb4ce1b520e5aa1cef26d0762c538ece5892c0
Status: Downloaded newer image for python:latest
docker.io/library/python:latest
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp>
```

By default chooses latest. Otherwise provide **:version_number.**

To check available images:

```
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker image ls

IMAGE            ID             DISK USAGE   CONTENT SIZE   EXTRA
mywebapp:01      89eacf391531       2.11GB          491MB    U
mywebapp:02      b90c1a14ca28       2.11GB          491MB
python:latest    0040600478fe       1.63GB          432MB
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp>
```

Now let's try to run this image:

```
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker run python:latest
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker image ls

IMAGE            ID             DISK USAGE   CONTENT SIZE   EXTRA
mywebapp:01      89eacf391531       2.11GB          491MB    U
mywebapp:02      b90c1a14ca28       2.11GB          491MB
python:latest    0040600478fe       1.63GB          432MB    U
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

```
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS                     PORTS
a2a1314b33eb   python:latest   "python3"                36 seconds ago   Exited (0) 35 seconds ago
d36ecc825040   2ca300af969a    "docker-entrypoint.s…"   42 hours ago     Exited (255) 16 hours ago   0.0.0.
48ff84ea22fc   2ca300af969a    "docker-entrypoint.s…"   42 hours ago     Exited (255) 16 hours ago   0.0.0.
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp>
```

```
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED
a2a1314b33eb   python:latest   "python3"                36 seconds ago
d36ecc825040   2ca300af969a    "docker-entrypoint.s…"   42 hours ago
48ff84ea22fc   2ca300af969a    "docker-entrypoint.s…"   42 hours ago
```

```
app> docker ps -a
 CREATED          STATUS                      PORTS                    NAMES
 36 seconds ago   Exited (0) 35 seconds ago                            gifted_merkle
 42 hours ago     Exited (255) 16 hours ago   0.0.0.0:3002->3000/tcp   infallible_haslett
 42 hours ago     Exited (255) 16 hours ago   0.0.0.0:3000->3000/tcp   charming_yalow
```

Python itself a software, since we have not created any python file, then there's no need of python. That is the reason why nothing happened after running python image. In,

```
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker run python:latest
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker image ls

IMAGE            ID             DISK USAGE   CONTENT SIZE   EXTRA
mywebapp:01      89eacf391531       2.11GB          491MB    U
mywebapp:02      b90c1a14ca28       2.11GB          491MB
python:latest    0040600478fe       1.63GB          432MB    U
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

In dockerhub, there are pre-defined images, where we can directly verify, Eg: web server, like NGINX, apache, etc.
Let's use NGINX,

```
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
9baba07a35b6: Pull complete
f0b77348d9b0: Pull complete
0289d65812c3: Pull complete
980067d12da2: Pull complete
4174e33a2c9e: Pull complete
6b40784e4837: Pull complete
ec781dee3f47: Pull complete
00238b7dc6b2: Download complete
c96ca1f4ddf7: Download complete
Digest: sha256:dec7a90bd0973b076832dc56933fe876bc014929e14b4ec4992395140537011
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp>
```

Nginx is a web server, it always runs in the background and waiting for the incoming request from the browser, whenever it access from the browser, nginx accepts the request, and represents the web page, nginx is completely used for deploying web app/applications. Nginx by default listens at port 80.
When we run nginx, meaning inside the container it listens at port 80. Similar to web application listening to port 3000.
In order to verify, need to do port binding.

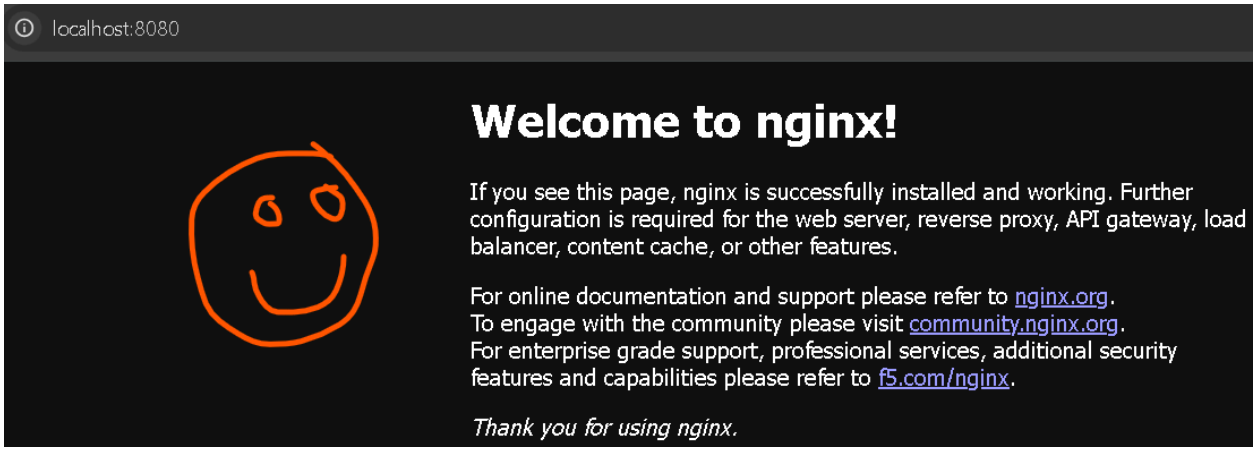
```
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker image ls

IMAGE            ID             DISK USAGE   CONTENT SIZE   EXTRA
mywebapp:01      89eacf391531       2.11GB          491MB    U
mywebapp:02      b90c1a14ca28       2.11GB          491MB
nginx:latest     dec7a90bd097        240MB         65.8MB
python:latest    0040600478fe       1.63GB          432MB    U
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp>
```

Lets run the nginx image,

```
PS C:\Users\kiran\OneDrive\Desktop\Web Application\testapp> docker run -p 8080:80 nginx:latest
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuratio
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/18 06:55:29 [notice] 1#1: using the "epoll" event method
2026/03/18 06:55:29 [notice] 1#1: nginx/1.29.6
2026/03/18 06:55:29 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
```

Mean while, visit: http://localhost:8080/



It continuously running at the background, and getting related logs:

```
2026/03/18 06:55:29 [notice] 1#1: start worker process 39
2026/03/18 06:55:29 [notice] 1#1: start worker process 40
2026/03/18 06:55:29 [notice] 1#1: start worker process 41
2026/03/18 06:55:29 [notice] 1#1: start worker process 42
2026/03/18 06:55:29 [notice] 1#1: start worker process 43
2026/03/18 06:55:29 [notice] 1#1: start worker process 44
172.17.0.1 - - [18/Mar/2026:06:58:06 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gec
ko) Chrome/146.0.0.0 Safari/537.36" "-"
172.17.0.1 - - [18/Mar/2026:06:58:07 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) App
leWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36" "-"
2026/03/18 06:58:07 [error] 29#29: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.17.0.1, server: loc
alhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost:8080", referrer: "http://localhost:8080/"
```